# Exercise 1

May 25, 2026

## 1 Exercise 1

Exercise 1: Build a Signal object

[1]:
```python
import numpy as np
import matplotlib.pyplot as plt
```

[2]:
```python
class Signal:

    def __init__(self, tt, xt):
        self.tt = np.array(tt)
        self.xt = np.array(xt)
        self.fs = (self.tt.size-1)/(self.tt[-1]-self.tt[0])

    def duration(self):
        return self.tt.size/self.fs

    def energy(self):
        return np.sum(np.abs(self.xt)**2)/self.fs

    def plot(self):

        fig = plt.figure()
        ax1, ax2 = fig.subplots(2, 1, sharex=True)
        ax1.plot(self.tt, np.real(self.xt), '-b')
        ax1.set_ylabel('Re[x(t)]')
        ax1.grid(alpha=0.6)
        ax1.set_title(f'Signal $x(t)$, $f_s$={self.fs:.1f} Hz')
        ax2.plot(self.tt, np.imag(self.xt), '-r')
        ax2.set_ylabel('Im[x(t)]')
        ax2.set_xlabel('t [sec]')
        ax2.grid(alpha=0.6)

        plt.show()
```

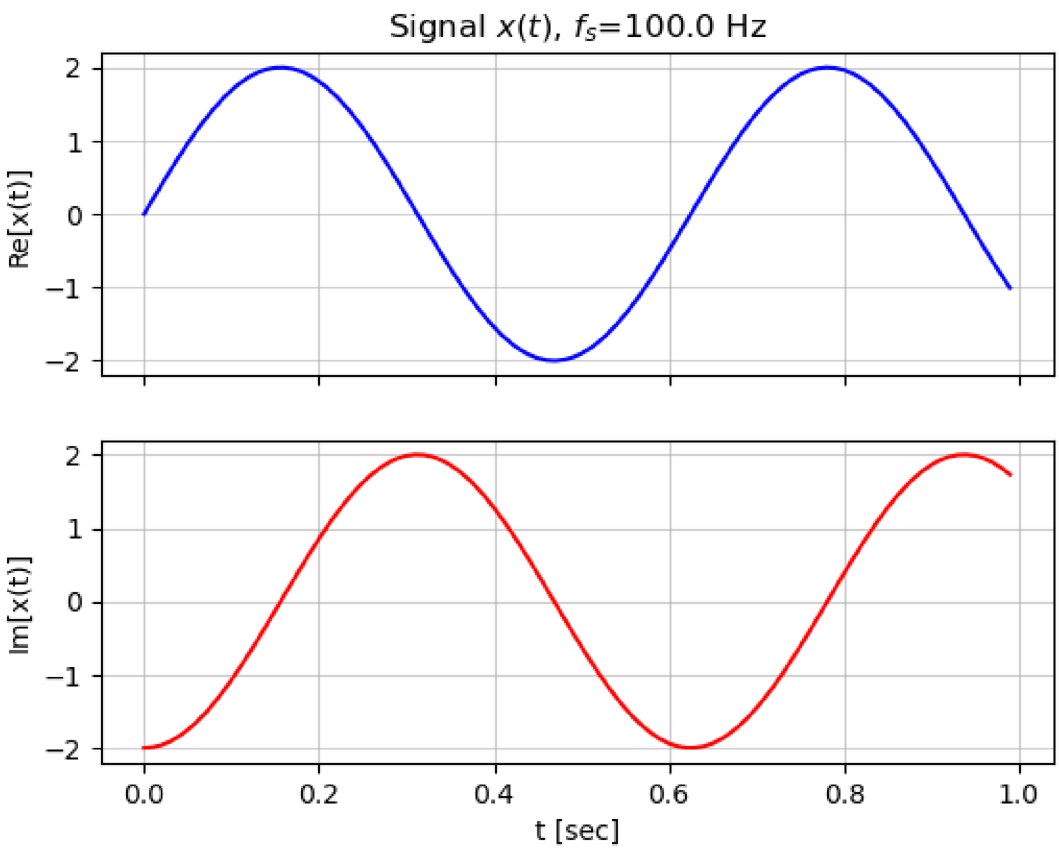
[3]:
```python
fs = 100
tlen = 1
tt = np.arange(int(np.round(tlen*fs)))/fs
```

```
A1, f1, th1 = 2, 1.6, -90
x1t = Signal(tt, A1*np.exp(1j*(2*np.pi*f1*tt+np.pi/180*th1)))
```

[4]:
```
print(x1t.fs)
print(x1t.duration())
print(x1t.energy())
```

```
100.0
1.0
4.0
```

[5]:
```
x1t.plot()
```




### Thinking exercise

- Why is fs stored?
- Why is energy() inside the object?
- Should plotting belong here?
- What should remain external?

### Answers

- fs relates the samples xt to the time axis.
- Energy is a property of the signal xt.
- A plot of signal xt is another representation of the signal. It is debatable whether that should be done internally or externally.
- The use of this signal in a larger system, e.g., an AM communication system.

---

[ ]:
```

```